

1. CRISP-DM Template for SDG Dashboard Design

1. Business Understanding

SDG Goal: 14 – Life below water

Target: 14.6 - End subsidies contributing to overfishing and illegal fishing

SDG Indicator: 14.6.1 - Combat illegal, unreported and unregulated fishing

Introduction:

Illegal, unreported, and unregulated (IUU) fishing, coupled with overfishing, is a significant driver of marine biodiversity loss in Spain and the Mediterranean Sea and one of the target that the SDG Goal 14 looks to solve during this decade. The exploitation of marine resources has intensified the decline of critical fish populations, destabilized ecosystems and threatening long-term sustainability. Although Spain has adopted international and regional regulations to combat IUU fishing, challenges remain in the implementation and enforcement of these laws. This research aims to explore the impacts of IUU fishing on marine biodiversity while evaluating Spain's progress in enforcing policies to curb overfishing. Understanding the connection between IUU practices and biodiversity loss is key to improving marine conservation and ensuring effective policy action in the Mediterranean region.

Problem Statement:

Illegal, unreported, and unregulated (IUU) fishing and overfishing continue to threaten marine biodiversity in Spain and the surrounding Mediterranean Sea, undermining conservation efforts. Despite regulatory frameworks, the implementation and enforcement of IUU laws remain challenging, hindering progress in protecting marine ecosystems.

Research Questions:

How does illegal fishing and overfishing (IUU) affect the biodiversity of marine life in Spain and its surrounding waters and what progress has Spain made in implementing IUU regulations?

2. Data Understanding

Sources of the data:

1. FAO – Food and Agriculture Organization of the United Nations

The data I have gathered from this source is mostly related to Global Capture Production values for species and Global Aquaculture production quantity. I have used the filters they have available to search for data related to my country of study, which is Spain, as well as

for the time period 2018-2022. The choice of this time period is mostly due to the multiple sources and the abundance of data available during these specific years. The other use of this database was for researching each species I was interested in and pulling out information regarding global capture values and aquaculture production values. The variables I have chosen from this database are: Year (2018-2022), Country name (Spain), Unit name (Tonnes), Aquaculture Production Value, Global Capture Value, AP value (each year), GC value (each year). The data is being integrated into the spreadsheet I have manually created, titled **"Research data IUU."**

Sources:

- <https://www.fao.org/fishery/en/aqspecies/search>
- https://www.fao.org/fishery/statistics-query/en/aquaculture/aquaculture_quantity
- https://www.fao.org/fishery/statistics-query/en/capture/capture_quantity

2. IUU Fishing Index

The database provided by the IUU Fishing Index is going to take a significant part of my dashboard, as this is the website where I will retrieve data related to Spain's law implementation regarding IUU regulation. The variables that I will pull from this website are related to indices and scores, such as Coastal, Flag, Port & General Score, and their indices: Vulnerability, Prevalence, and Response. The available data covers the years 2019, 2021, and 2023, as the reports are usually published every two years.

The data is being integrated into the spreadsheet I have manually created, titled **"IUU Index Score."**

Sources:

- <https://iuufishingindex.net/profile/spain>

3. IUCN Red List

The part related to marine biodiversity in my research question will be supported with data from the IUCN Red List Database, where I can pull variables such as species' common and scientific names, conservation actions (whether Spain is implementing conservation actions to protect these species), species population trends (whether they are increasing or decreasing), and the last time these species were assessed regarding their endangered status. Also, the site have been used to gather data regarding the number of species for each endangered status.

The data is being integrated into the spreadsheet I have manually created, titled **"Research data IUU."**

Sources:

- <https://www.iucnredlist.org/search/stats?redListCategory=en>

4. Other sources

The sources listed below have been integrated indirectly into my dashboard and research, either serving as a basis for further research or as guides to observe trends between variables.

- [Spain – A progress report on a decade of combating IUU fishing: an update from the 2020/2021 biennial report \(pdf file\)](#)
 - <https://oap.ospar.org/en/ospar-assessments/committee-assessments/biodiversity-committee/status-ospar-network-marine-protected-areas/assessment-sheets-mpa/>
 - <https://globalinitiative.net/wp-content/uploads/2023/11/The-Illegal-Unreported-and-Unregulated-Fishing-Risk-Index-GITOC-Poseidon-DECEMBER-2023.pdf>
- Regarding the quality of the data, I have been able to find most of the information I was looking for and that I want to include in my research and my dashboard. I wanted to show the evolution of the population of different marine species over the chosen period (2018-2022), but I found that it is very difficult to provide an exact number or even estimate it. Therefore, I decided to drop that idea and instead search for information on population trends and conservation actions. I expected to find much more data regarding this topic, but I was surprised that you really need to clarify things yourself in order to actually show a relation between two variables.

3. Data Preparation

The **Research data IUU Excel spreadsheet** was created from scratch using various sources to select only the data relevant to my research. Spain is the target country, and all of the data is connected to it.

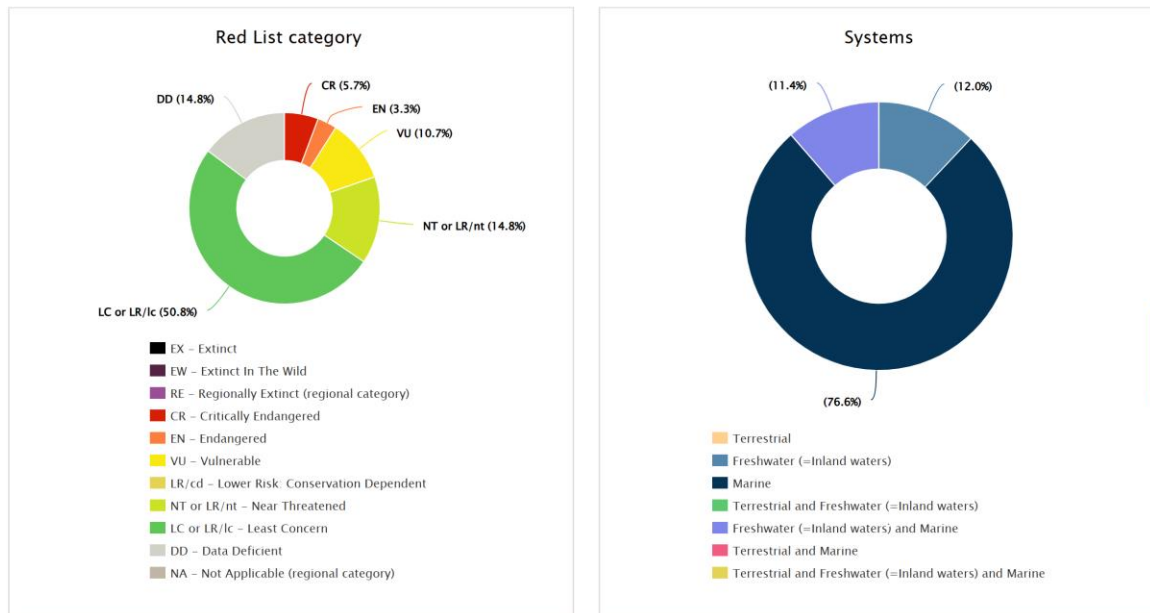
The *Aquaculture Production* and *Global Capture Spain files* were created by pulling data from the FAO Statistics Query official site, selecting only the data for the years 2018-2022, with the year 2022 being the most recent value they provide for these variables.

The *Red List Index sheet* contains informations regarding the number of species for each of the endangered status available on the site (source: <https://www.iucnredlist.org/search/stats?redListCategory=en>) . I gathered the data by using the filters from the website and selecting the area I was interested in, as you can see down below.

The screenshot shows the IUCN Red List search interface. The top navigation bar includes links for '2024-2', 'Login / Register', 'What's New', 'Contact', 'Terms of Use', and 'English'. The main search bar contains the text 'Names - common, scientific, regions etc...'. Below the search bar, there are tabs for 'Grid', 'List', 'Map', and 'Stats', with 'Stats' currently selected. A 'Relevance' dropdown menu is also visible.

On the left side, there are filters for 'TYPE' (Species, Featured Regions, Documents, Red List Indices) and 'SEARCH FILTERS' (Taxonomy, Red List Category, Land Regions, Country Legends, Marine Regions, Threats). The 'Red List Category' filter is expanded, showing 11 categories. The 'Land Regions' filter is set to 'Spain'. The 'Threats' filter is set to '2 - Agriculture & aquaculture' and '6 - Human intrusions & disturbance'. The 'Habitats' filter is set to '10 - Marine Oceanic', '11 - Marine Deep Benthic', '12 - Marine Intertidal', '13 - Marine Coastal/Supratidal', '15 - Artificial/Aquatic & Marine', '5 - Wetlands (inland)', and '9 - Marine Neritic'. The 'Systems' filter is set to 'Freshwater (=Inland waters)', 'Freshwater (=Inland waters) and Marine', and 'Marine'. The 'Marine Regions' filter is set to 'Mediterranean and Black Sea'. The 'Include' filter is set to 'Species'.

The main content area displays 'RESULTS (122)' and a list of Red List categories with their corresponding counts: CR - Critically Endangered (X), DD - Data Deficient (X), EN - Endangered (X), EW - Extinct In The Wild (X), EX - Extinct (X), LC or LR/lc - Least Concern (X), LR/cd - Lower Risk: Conservation Dependent (X), NA - Not Applicable (regional category) (X), NT or LR/nt - Near Threatened (X), RE - Regionally Extinct (regional category) (X), and VU - Vulnerable (X).



The *Species sheet* was created by researching various sources on endangered or vulnerable marine species from Spain and its surrounding waters, with their endangered status being the most recent one provided on the IUCN Red List of Threatened Species (URL: <https://www.iucnredlist.org/search/stats?redListCategory=en>), as well as their population trends and conservation actions.

*For the specific *species tables* (e.g., *Anguilla anguilla*, *Squatina squatina*, *Lamna nasus*, etc.), I've taken the values for the years 2018-2022 of the variables Aquaculture Production and Global Capture Production from the FAO Aquatic Species official website (URL: <https://www.fao.org/fishery/en/aqspecies/search>) by searching for the specific endangered species I've found previously on the IUCN Red List.

(*)As an update for this table, I have decided to not use them directly anymore in my Dashboard, but also not fully remove them since they provided an useful insight for my research and a clear pathway towards the improvement of my project. I have decided to hide them in Excel and remove them from Power BI.(*)

The *species GCP sum 2018-2022 sheet* summarizes the overall global capture production values for the species being analyzed during this research. I performed a summarization using the SUM formula and added two new variables: Total per species (shows the total GCP value from 2018-2022 for a specific specie) and Total per year (shows the total GCP value in each year for all of the species that are being documented during this research). All of the numeric cells (besides the year columns) have been formatted in the number category as decimal numbers with two decimal places and to use 1000 separators. The species AP sum 2018-2022 went through the same process.

For the **IUU Index Score spreadsheet**, I gathered the data from the website <https://iuufishingindex.net/profile/spain> and its corresponding downloadable Excel/CSV file and created a separate table to visualize the data better. Prior to this step, in the Excel file provided on the site, I applied some data manipulation and preparation techniques, such as filtering the columns to show only data on Spain and/or different scores (General/Flag/Port Coastal). I also tested by filtering only for the *Mediterranean and Black Sea*, but I decided to include all of the oceans available since they provided a more profound understanding of my research and had a clear connection with the general score.

The **IUU fishing index score spreadsheet** contains the same data from the *data file for IUU Fishing Risk Index scores* (source: <https://iuufishingindex.net/data-files>), but I filtered the columns to show me the relevant data I aimed for: Deleted the *indicator ID* column, filtered the *country* column to show only *Spain*, and for each page in my newly created spreadsheet, I filtered for the indicator I wanted to show (general, flag, port, coastal) for all of the years available (2019, 2021, 2023).

For the **IUU Fishing Incidents vs. Marine Protected Area (MPAs) Coverage spreadsheet** that contains the IUU reported incidents and MPA Coverage I've analyze multiple sources and data bases and couldn't be able to choose one that it is general available for my research.

4. Modeling

For the **IUU Index Score** during my input of data in Power BI, I decided to unpivot the columns representing specific years (2019, 2021, 2023) for each sheet (Flag, Port, Coastal, and General, along with their corresponding indicators – Vulnerability, Prevalence, Response) in order to display the variables appropriately on my radar charts, my visual of choice to show the evolution of the score for each indicator over the reported years. I made some changes to the data type to ensure that variables such as Year and Score are set as whole numbers instead of text.

For the **IUU fishing index score sheets**(*Flag 2019-2023, Coastal 2019-2023, Port 2019-2023, General 2019-2023*) I deleted the blank rows and columns generated by Power BI during data input, using the *Remove Blank Rows option* in order to avoid overload or calculation errors.

For *IUU incidents sheet*, I changed the data type of the MPA Coverage from percentage to decimals for more accurate representation in the *Correlation between Reported IUU Fishing Incidents and MPA Coverage by Year and Region* scatter plot. For the *Predictive Analysis* visual I've decided to change again the data type back to percentage.

For all of the pages of my Dashboard, I've added Page Navigators on each of them to be able to make the navigation smoother.

Before introducing and presenting each of the visuals, I will summarize the reason of input for each page and how it is relevant to the main research question.

Visualizations:

1. Species and General Data

- *Species informations:*

Description: This visual is relevant towards the first part of the research question which touch the topic of how marine life is impacted by these kind of activities. The table shows how different species from marine environments close or within Spain region are being affected, pointing in particular they endangered status and their population trends, but also if there are any conservation actions that are currently being applied in order to stop those decreasing trends. This is also related to the second part of the research that want to bring up what are the actions that Spain is actively taking in order to reach the SDG Goal 14.

Visual: Table (Columns: Common Name, Scientific Name, Endangered Status, Conservation Actions, Most recent assessed Year); can be interacted with and display using the *Species Informations* button.

Changes: Changed the title, changed the font to Candara; added three slicers

Data source: Species spreadsheet from **Research data IUU**

- *Sort by Endangered Status, Sort by Conservation Actions, Sort by Year*

Description: These visuals are being placed to help with the navigation within the table *Species informations* that I have introduce above.

Visuals: Slicers(each for their category: endangered status, conservation actions and most recent assessed year); can be interacted with and display using the *Species Informations* button.

Data source: Species spreadsheet from **Research data IUU**

- *Red List Category for Spain*

Description: The role of this visual is to show the dispersion of species over the endangered statuses that have been assessed by IUCN. It is clearly correlated with the first part of my research question regarding the effect that IUU has over marine biodiversity and environments, this being the reason of choosing *Agriculture and Aquaculture* and *Human instruction and distrupction* on the Threats filter.

Visual: Donut chart(Legend: Endangered Status; Values: Max of Number of endangered species); can be interacted with and display using the *Red List Index* button.

Changes: Changed the color of the slices and rotated the visual in order to display all of the values names.

Data Source: Red List Index from **Research data IUU**

- *Correlation between Reported IUU Fishing Incidents and MPA Coverage by Year and Region:*

Description: This visual plays a key role in my research and is particularly intriguing to analyze. As I will highlight further in the Evaluation section, it reveals a correlation between two variables that might initially be expected to have little or no relationship. I aimed to demonstrate the positive impact of taking concrete steps to protect marine areas. The increased implementation of regulations in these regions

helps detect illegal and unregulated fishing activities more effectively. However, this doesn't necessarily mean there is a rise in the number of these incidents, but rather an improvement in monitoring and enforcement.

Visual: Scatter Chart(Values: Year, X-axis: Sum of IUU Fishing Incidents, Y-axis: Sum of MPA Coverage(%), Legend: Region)

Changes: unpivoted the columns for the years to make the tabel more manageable for the calculations before inputting in Power BI; changed the data type of the MPA Coverage(%) from percentes to decimals for more accurate representation for the visual; decided to keep the standard colors for the Legend(Regions)

Data source: IUU incidents from **IUU Fishing Incidents vs. Marine Protected Area (MPAs) Coverage**

- *Predictive Analysis of Reported IUU Fishing Incidents by MPA Coverage growth:*
Description: By incorporating a forecasting visualization, I aimed to highlight the correlation between these two variables from a future perspective. Analyzing past trends over the years shows growth in both MPA coverage and reported IUU fishing incidents, which, theoretically, suggests that this trend may continue. While other factors—such as legal actions, new laws, emerging technologies, or socioeconomic influences—can certainly affect the outcome, the forecast provides a valuable guide for visualizing future trends based on the data currently available to us.
Visual: Line Chart(X-Axis: MPA Coverage(%), Y-axis: Average of IUU Fishing Incidents)
Changes: Activated the Forecast option and kept it at a 95% confidence interval; changed the color of the Confidence Shade
Data Source: IUU incidents from **IUU Fishing Incidents vs. Marine Protected Area(MPAs) Coverage**
- *Correlation Coefficient for IUU Fishing Incidents and MPA Coverage:*
Description: This card displays the value after calculating the correlation coefficient, which shows a result of 0.89 that indicates a strong correlation between these two variables.
Visual: Card (Fields: IUU Fishing Incidents and MPA Coverage (%) correlation for Year); can be interacted with and display using the *Correlation Coefficient* button.
Changes: Created a new measure for the correlation coefficient using the Quick measure option on Power BI that generates a DAX formula by inputing the variables: IUU Fishing Incidents, MPA Coverage (%) and Year; Input the new measure into a Card Visual and add a Title, hide the Category label and add a border for design.
Data Source: IUU incidents from **IUU Fishing Incidents vs. Marine Protected Area (MPAs) Coverage**

2. Aquaculture & Global Capture

Page description: The visuals on this page highlight two key categories related to the fishing industry. Global Capture refers to the harvesting of wild fish and marine organisms from natural habitats such as oceans, seas, rivers, and lakes. This method

involves traditional fishing techniques like trawling, longlining, and netting, and depends on the availability of wild fish populations. On the other hand, aquaculture, or fish farming, is the practice of breeding, raising, and harvesting aquatic organisms in controlled environments like ponds, tanks, or coastal areas. The visuals are mainly used to contrast these two approaches, clearly showing the higher values of global capture compared to the relatively lower values of aquaculture. This suggests a lack of controlled environments for fish farming or possibly limited resources to expand such spaces.

- Global Capture Value in Spain during 2018-2022:*
Visual: Stacked column chart(X-axis: Year; Y-axis:Max of Global Capture Value)
Changes: changed the Title of the Y-axis, added a Zoom slider for easy control and flexibility in seeing clear the values for each year, or for comparing
Data Source: Aquaculture production from **Research data IUU**
- Aquaculture Value in Spain during 2018-2022:*
Visual: Stacked column chart(X-axis: Year; Y-axis: Max of Aquaculture value)
Changes: changed the Title of the Y-axis, added a Zoom slider for easy control and flexibility in seeing clear the values for each year, or for comparing
Data Source: Global Capture Spain from **Research data IUU**

3. Coastal Score

Page 3-6 description: As they are similar in structure and the kind of information that it is being presented, I will include them all in one summarizing description. The IUU Fishing Index uses coastal, general, port, and flag scores to assess how countries manage illegal fishing. The Coastal Score evaluates the impact of IUU fishing in a country's waters, while the Port Score looks at how well a country monitors its ports to prevent illegally caught fish from entering the market. The Flag Score measures a country's control over vessels registered under its flag, even when they fish internationally. The General Score combines these elements to give an overall view of a country's effectiveness in combating IUU fishing. Together, these scores provide insights into the country's ability to manage illegal fishing at multiple levels. This pages are being more related to the second part of the research question by showing the action that are being taken by Spain. It's important to note that the data shown covers only the years 2019, 2021, and 2023, not as a result of a selection preference, but because these are the years when the reports were made and published, typically on a two-year cycle.

- Coastal Score by Prevalence, Response, Vulnerability:*
Visuals: Three separate Radar Charts for each indicator (Prevalence, Response, Vulnerability) (Category: Indicator Name – each from their corresponding table, Y axis: Score – each from their corresponding table)

Changes: Created relationships between the Year column from each table for further use in the other visuals. Unpivoted the columns for Year and Score for each table; choose for the Y axis the Sum of Scores instead the Count of them, as the system auto filled, to display the exact values for each indicator: turned off Legend and Draw Lines; changes done to design, such as colors, opacity on different colors; changed the title;
Data Source: Coastal Score by Prevelence, Response, Vulnerability from **IUU Index Score**

- *Select the Year:*
Visual: Slicer (Field:Year)
Changes: Slicer setting: Tile; added Effects such as Background colour;
Data Source: Coastal by Vulnerability from **IUU Index Score**
- *Mean of Coastal Score:*
Description: The use of this card visual is to show the progress towards hitting a certain goal made by Spain while implementing IUU Fishing regulations. The goal in our case is to get to a score as close as possible or even equal 1, as the methodology behind states that a score equal to 1 shows a certain improvement and a score of 5 shows the need of improvement. The card is interactive with the slicer for Years make it more easily to see if Spain is making progress related to the indicator (whether is Coastal Score, or the other indicators that are being shown – General, Port and Flag).
Visual: Card (Fields: Mean_Coastal_Score)
Changes: Created a New Measure for the Mean using the DAX formula: Mean_Coastal_Score = AVERAGE('Coastal 2019-2023'[Score]); added a border to the card and changed the name of the visual variable;
Data Source: Coastal 2019-2023 from **IUU fishing index score**
- 4. **Flag Score** – the methods, techniques and design used on this page are identical to the ones from Coastal Score page, as they display similar concepts, but within different parameters.
- 5. **Port Score** – the methods, techniques and design used on this page are identical to the ones from Coastal Score page, as they display similar concepts, but within different parameters.
- 6. **General Score** – the methods, techniques and design used on this page are identical to the ones from Coastal Score page, as they display similar concepts, but within different parameters.

5. Evaluation

After an in-depth analysis and a final overview of the dashboard I've come to the conclusion that in its fullness covers the points I've wanted to show and which I have presented in my business objectives. The Dashboard presents a comprehensive overview of the chosen SDG Goal(Goal 14) by addressing the impacts of IUU fishing and overfishing on marine biodiversity, but also pointing the evolution of Spain and over which aspects this country still needs further improvement or immediate attention(for example, the growth of Port

Score between the report from 2019 and the one from 2023, showing a decline in actions taken towards reaching the initial objective, which is a score as closer to 1 as possible).

The dashboard has a user-friendly navigation system for which I implemented page navigators to favorize a smooth navigation over the project. In addition, the dashboard displays a comprehensive data representation by utilizing different visuals such as sccttaer plots, radar charts, and line charts, offering multiple perspectives on the data.

While the dashboard effectively visualizes data, it may not fully capture the underlying complexities of how IUU fishing impacts marine biodiversity. These relationships can be influenced by various external factors, such as climate change and advancements in fishing technology. Additionally, navigating the available data on this specific topic can be challenging due to multiple factors that are often overlooked or deemed irrelevant for public discourse. This is especially true in cases involving illegal activities, where data is frequently missing—either due to limitations on public disclosure or because certain information has yet to be discovered. A clear example of this issue is evident on the first page of the dashboard, particularly in the scatter plot illustrating the correlation between reported IUU fishing incidents and MPA coverage. The scatter plot indicates that while the growth of protected areas correlates with an increase in reported IUU incidents, this does not necessarily mean that the occurrences of these incidents are rising. Rather, the enforcement of more protective measures in these areas is likely leading to a higher detection rate, making it easier for authorities to identify and report such incidents.

I think the dashboard could be improved by incorporating additional data sources, such as socioeconomic factors or stakeholder perspectives, to provide a more comprehensive view of the impact of IUU fishing on marine biodiversity. Enhancing interactivity with features like drill-down capabilities and tooltips would engage users further and allow them to explore specific data points in greater detail. Implementing advanced analytical techniques, such as machine learning models or statistical tests, could uncover deeper insights and strengthen the predictive capabilities of the dashboard. Additionally, establishing a system for regularly updating the data and visuals would help maintain the dashboard's relevance, ensuring that stakeholders have the most current information for informed decision-making in marine conservation efforts.

6. Deployment

To deploy the dashboard, I'll start by publishing it to the Power BI service so that it's available online. I'll share access links and invite these stakeholders to check it out directly in Power BI, providing some tips on how to navigate and understand the visuals.

My university colleagues and the SDG team from our university are the target audience for my dashboard. Presenting the dashboard to them will provide valuable information about the impacts of IUU fishing and marine biodiversity, and I hope it will spark their interest in these important issues. Engaging with this audience will also allow for discussions that could lead to new ideas and insights, enhancing the overall understanding of the topic.

Since my databases are manually created from different sources, I'll ensure the dashboard stays up to date by implementing a systematic approach. First, I'll create a schedule for myself to regularly review and update the data, perhaps monthly or quarterly, depending on how often new information becomes available. I'll keep track of which sources need to be checked and set reminders to update the database with any new data I find. Additionally, I can document the process for updating the data so it's clear and easy to follow in the future. If possible, I might also reach out to some of the data sources to see if they offer automated feeds or alerts for new information, which would help streamline the update process. By being organized and proactive about checking and updating the data, I can help ensure that the dashboard remains relevant and accurate.

7. Future Research

In order to overcome the limitations in my research, I could take several measures. Initially, I could examine the availability of more recent data sources concerning IUU fishing and marine biodiversity to ensure that the research is grounded in current information. Partnering with organizations focused on marine conservation could potentially facilitate access to more comprehensive datasets. Furthermore, I could explore emerging data analysis techniques or tools, like machine learning algorithms, that could offer more profound insights into the data's trends and patterns.

In terms of the audience, my dashboard is intended for my university peers and the SDG team. However, I could also consider connecting with local fishing communities, policymakers, and environmental organizations. By distributing the results to these groups, the research could help shape decision-making and increase awareness about the effects of IUU fishing on marine biodiversity. This involvement could also promote cooperative endeavors to promote sustainable fishing methods and enhance policies that benefit both the environment and the communities dependent on marine resources.